

Java Programming Brush up Tutorials

Day 2 Agenda:

- Discuss about Syntax of Basic Hello World Program
- Comments
- List and Types of data types
- Variables and its limitations
- Keywords/Reserved words

Basic Syntax of Java class

Syntax:

```
public class ClassName{  
    public static void main(String args[]){  
        //list of statements.....  
    }  
}
```

Explanation of each and every keyword:

public – Access specifier, whatever the code we are writing in this class, anyone can access through your directory and inherit the same another package as well (Will learn in future modules)

class – Every class starts with the keyword called class, means a specific group of items belongs to a particular one.

Example: Honda, Hero, Bajaj, Yamaha,.... All these are objects but all these objects belong to one group that is Bike Class/group.

void- Return type, here compiler expects something from the class once execution is finished till the end of the code. If we give void means nothing, so that it will never expect something from the class

main- For every program starts compiling with main function, here main is very important to start compile the code from line 1.

Example: Like our house how the main door is important likewise here in programming to enter the compiler into the program the main function is important.

String – It is a primitive data type that arguments list should be a type of String. Here string means a list of characters which are included in double quotes nothing but a String. Example: “Hello World”, “Welcome to Java Programming”

args[] – the array of arguments that we can give at the time of executing the code.

Explanation of Hello World!!! Program:

```
class HelloWorld{  
    public static void main (String args[]){  
        System.out.println(“Hello World!!!”);  
    }  
}
```

```
}  
  
}
```

As per the sytanx:

Class starts with name class

Here the new statement is **System.out.println("Hello World!!!");**

As our main aim is to print Hello World!!! on the screen, we should use an output function as per java, Here the output function is System.out.println();

Make sure every statement ends with semicolon (;)

Here System means the Computer/Laptop

out – output on the screen

println - Print the output with a new empty line after the statement.

System.out.println(); means print the output on the laptop screen with new line

Here the output is what we mentioned in the double quotes, as we mentioned Hello World!!! in the double quotes, that's the output.

Comments:

Comments are used to make our code to know others. For example, you are working in X company and developed one module as a part of product, and you got a good offer from some Y company thereby you left company X. Now company X has hired a new person to continue the work that you have left. Here to understand how you write the code for the one you did, to understand that by others you must mention the comments

For Example: `int a = 10; //Here a is an identifier which holds the value 10.`

There are two types of comments:

- Single Line comment (//)
- Multiline Comment (/*.....*/)

Example:

`//This is a single line comment`

Multiline Comment:

```
/*  
    Hai  
    Hello, Welcome to Java Programming  
    Thank you  
*/
```

List and Types of Data Types:

There are two types of data types:

- Primitive Data Types
- Non-Primitive Data Types

Primitive Data Types:

It is associated with variables and that variables can hold the specific type of value.

Data Type	Bits and Range
Byte	8 bits (-128 to 127)
Short	16 bits (-32,768 to 32,767)
Int	32 bits (-2,147,483,648 to 2,147,483,647)
Long	64 bits
Float	32 bits
Double	64 bits
Char	2 Bytes
Boolean	1 bit

Non-Primitive data types:

- Arrays
- Class
- Object
- String

Each non primitive data type is created by user whereas primitive data types are built in types and those come with java itself except String data type

Primitive data types hold some value, but non primitive data types are holds null value by default.

Refer the links for more info: https://www.w3schools.com/java/java_data_types.asp
<https://www.javatpoint.com/java-data-types>

Variables/identifiers and Its Limitations:

Variables are used to store values of respected data type.

For example: I would like to store value 10 of integer data type in variable a. Then here is the syntax: datatype variable_name = value;

Here in our case: **int a = 10;**

The limitations or the rules of variables/identifiers:

- Variable names can contain letters, digits, underscores, and dollar signs
- Variable names must begin with a letter
- Variable names should start with a lowercase letter, and cannot contain whitespace
- Variable names can also begin with \$ and _
- Variable names are case-sensitive ("myVar" and "myvar" are different variables)
- Reserved words (like Java keywords, such as int or Boolean) cannot be used as variable names

Simple program to execute all Primitive data types:

```
class Employee {  
    public static void main(String[] args) {  
        // Employee data  
        String employeeName = "Chandramouli Pyla";  
    }  
}
```

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```
int employeeID = 1231234;
int employeeAge = 23;
Long employeeNum = 2175120038L;
float employeeSalary = 6475.25f;
double employeeSalInRsPerYear = 6581444.1d;
boolean isLegal = true;

// Print variables
System.out.println("Employee name: " + employeeName);
System.out.println("Employee id: " + employeeID);
System.out.println("Employee age: " + employeeAge);
System.out.println("Employee Number: " + employeeNum);
System.out.println("Employee Salary: " + employeeSalary);
System.out.println("Employee Salary In Rupess per Year: " +
employeeSalInRsPerYear);
System.out.println("Is legal Employee or not: "+isLegal);
}
}
```